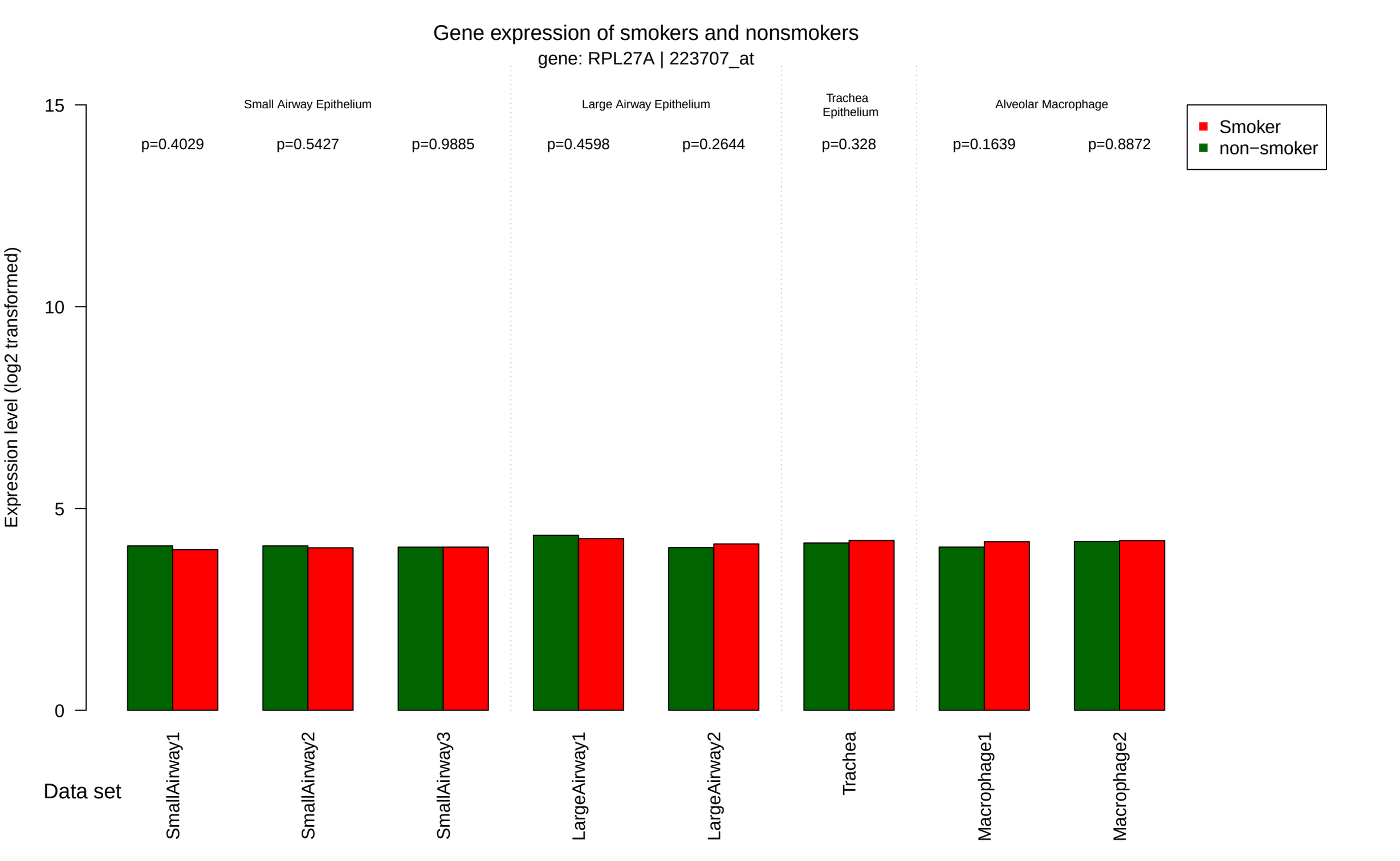
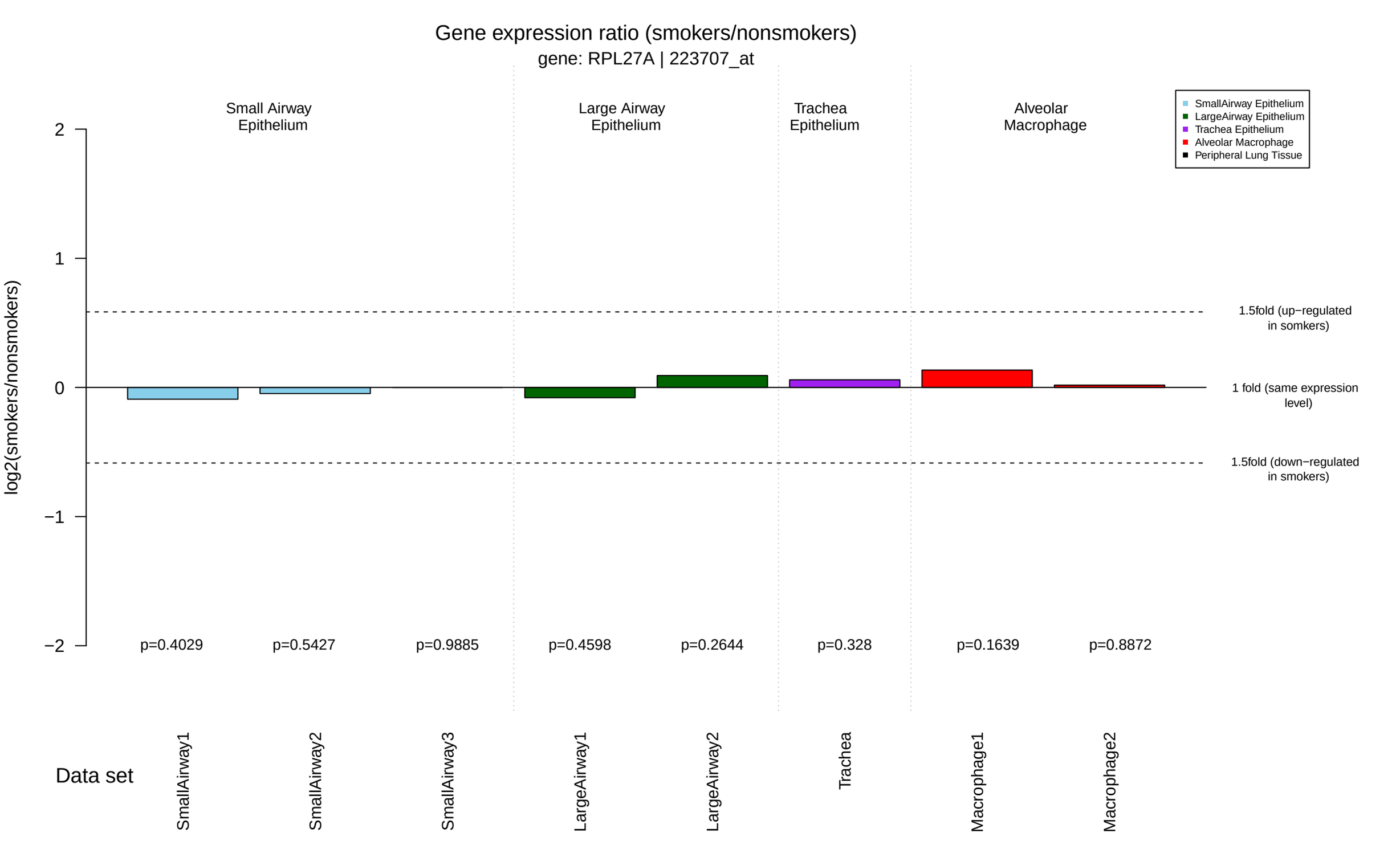
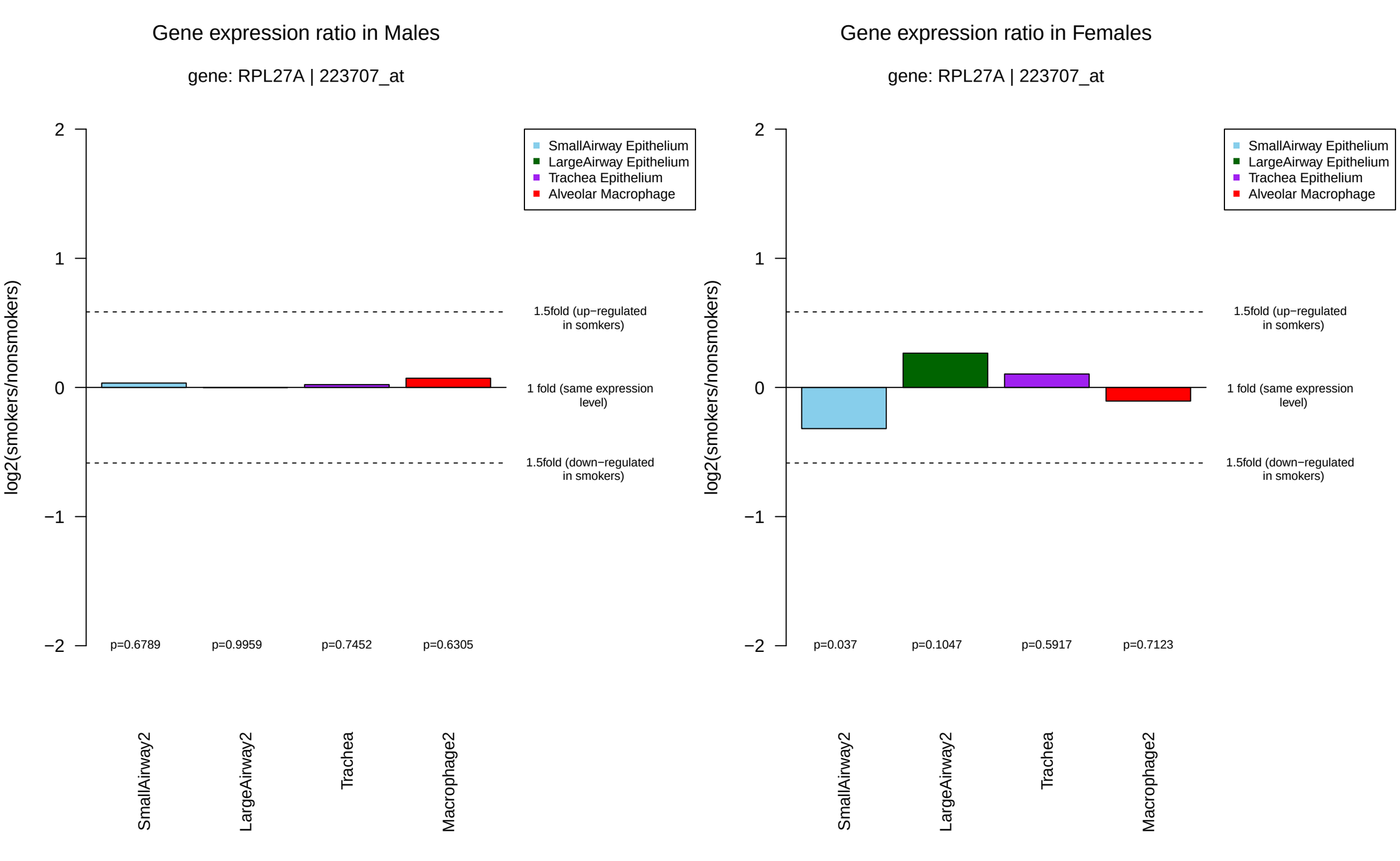
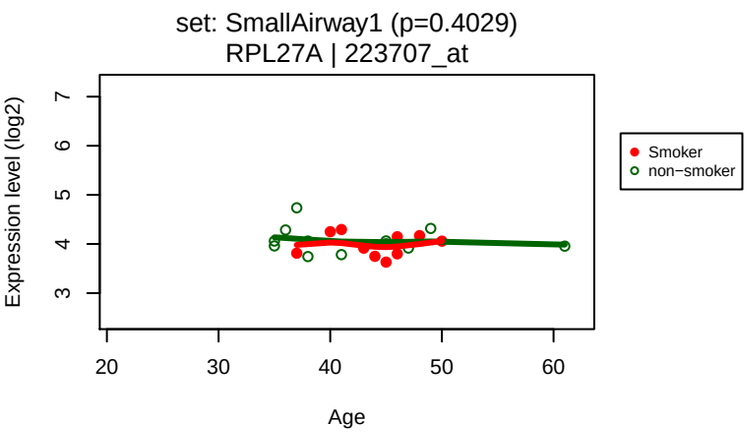
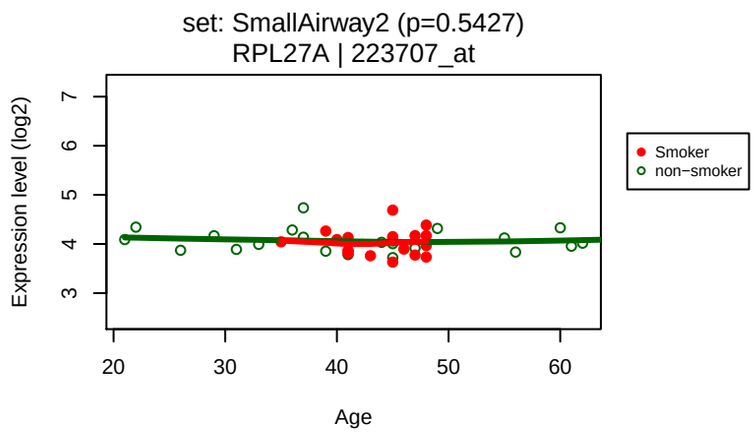
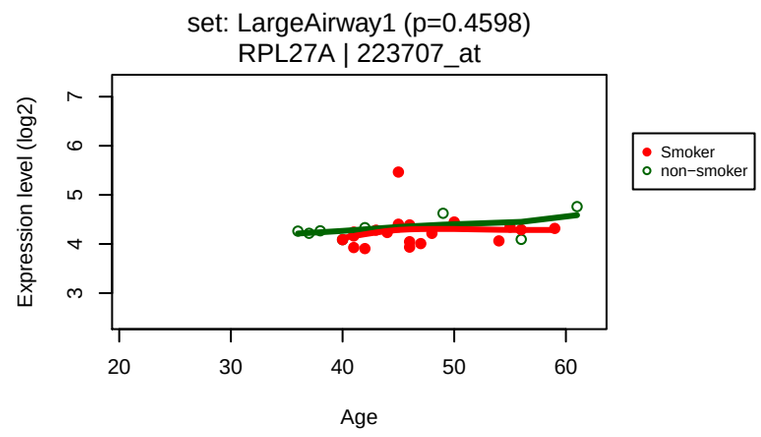

Gene expression of smokers and nonsmokers
gene: RPL27A | 223707_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4029
p=0.5427
p=0.9885
p=0.4598
p=0.2644
p=0.328
p=0.1639
p=0.8872
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

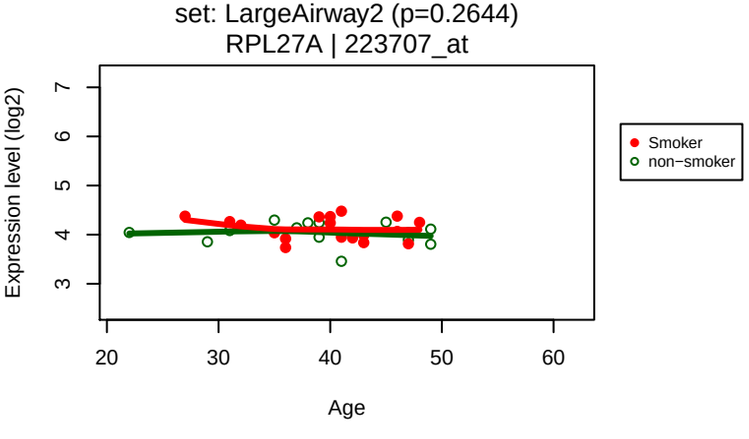

Gene expression ratio (smokers/nonsmokers)
gene: RPL27A | 223707_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4029
p=0.5427
p=0.9885
p=0.4598
p=0.2644
p=0.328
p=0.1639
p=0.8872
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

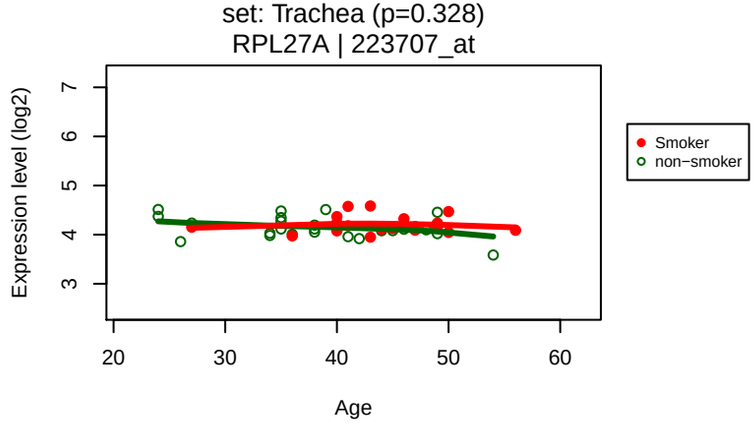

Gene expression ratio in Males
gene: RPL27A | 223707_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6789
p=0.9959
p=0.7452
p=0.6305
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPL27A | 223707_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.037
p=0.1047
p=0.5917
p=0.7123
SmallAirway2
LargeAirway2
Trachea
Macrophage2

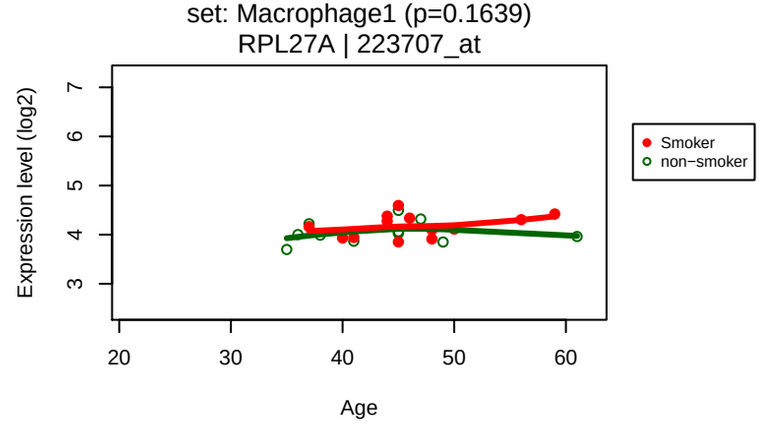

set: SmallAirway1 (p=0.4029)
RPL27A | 223707_at
Expression level (log2)
Age
Smoker
non−smoker

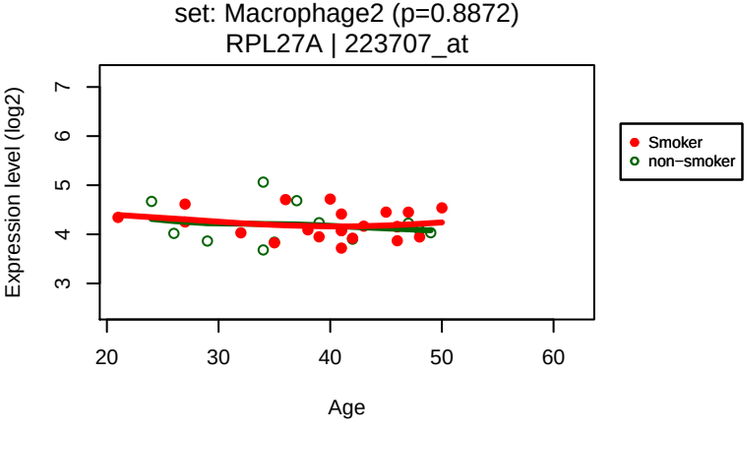

set: SmallAirway2 (p=0.5427)
RPL27A | 223707_at
Expression level (log2)
Age
Smoker
non−smoker

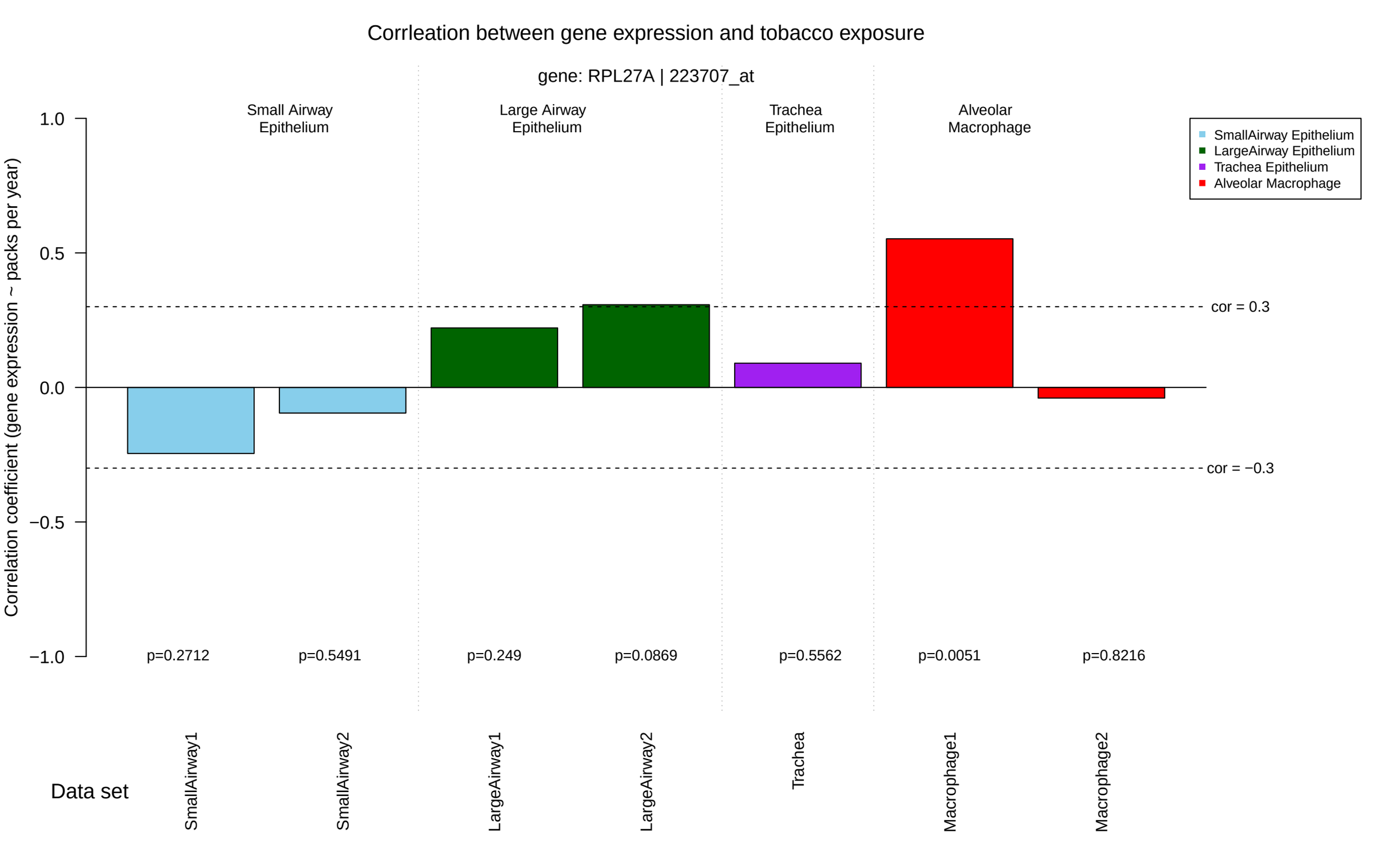

set: LargeAirway1 (p=0.4598)
RPL27A | 223707_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2644)
RPL27A | 223707_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.328)
RPL27A | 223707_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1639)
RPL27A | 223707_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8872)
RPL27A | 223707_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPL27A | 223707_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2712
p=0.5491
p=0.249
p=0.0869
p=0.5562
p=0.0051
p=0.8216
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2